A POSITIVE ANSWER TO THE COTYPE SUMMABILITY REPRESENTATION QUESTION

AGENT LANE 19

Source problem

The source paper is Lopez-Abad, Olmos-Prieto, and Uzcategui-Aylwin, *F_σ-ideals, colorings, and representation in Banach spaces*, arXiv:2501.15643. Question 3.9 asks whether an ideal that is $\mathcal{B}$-representable on a Banach space of non-trivial cotype q can be $\mathcal{B}$-represented, still in cotype q, by a sequence (x_n) for which $\sum_n \|x_n\|^q = \infty$.

$$\mathrm{Sum}((\|x_n\|)_n) \subseteq \mathcal{B}(\mathbf{x}) = \mathcal{C}(\mathbf{x}) \subseteq \mathrm{Sum}((\|x_n\|^q)_n)$$

for every sequence $\mathbf{x} = (x_n)_n$ in X. □

Question 3.9. *Suppose that an ideal is $\mathcal{B}$-representable on a space with non trivial cotype q. Is it possible to do it in such a way that the corresponding sequence $(x_n)_n$ satisfies that $(\|x_n\|^q)_n$ is not summable?*

4. Representing non-pathological F_σ ideals on spaces of continuous functions

Here ideals are understood in the convention used by the source and by Filipow–Tryba: they are proper ideals on $\mathbb{N}$ containing all finite subsets. The improper family $\mathcal{P}(\mathbb{N})$ is therefore outside the convention.

Definitions

For a sequence $\mathbf{x} = (x_n)$ in a Banach space X, write

$$\mathcal{B}_X(\mathbf{x}) = \left\{ A \subseteq \mathbb{N} : \sup_{F\subseteq A,\ F \text{ finite}} \left\| \sum_{n\in F} x_n \right\|_X < \infty \right\}.$$

For weights $w = (w_n)$ with $0 \le w_n < \infty$, write

$$\mathrm{Sum}(w) = \left\{ A \subseteq \mathbb{N} : \sum_{n\in A} w_n < \infty \right\}.$$

This summable ideal is non-trivial precisely when $\sum_n w_n = \infty$.

Proof Intuition

The source paper proves the one-sided cotype obstruction: if X has cotype q, then every $\mathcal{B}_X(\mathbf{x})$-set has finite $\sum \|x_n\|^q$. Thus a representation with divergent total $\sum_n \|x_n\|^q$ can exist only when the ideal sits inside some non-trivial summable ideal.

The converse is the useful trick. If $\mathcal{I} = \mathcal{B}_X(\mathbf{x}) \subseteq \mathrm{Sum}(w)$ and $\sum_n w_n = \infty$, add one orthogonal ℓ_q-coordinate with length $w_n^{1/q}$ to each vector:

$$y_n = (x_n, w_n^{1/q} e_n) \in X \oplus_q \ell_q.$$

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Finite sums then split exactly into the original X-partial sum and the weight sum. Hence the new bounded-partial-sum ideal is $\mathcal{B}_X(\mathbf{x}) \cap \mathrm{Sum}(w) = \mathcal{I}$, while the total q-mass diverges because of the added ℓ_q-coordinate.

Main theorem

Theorem 1. *Let $2 \le q < \infty$, and let $\mathcal{I} = \mathcal{B}_X((x_n))$ for a sequence in a Banach space X of cotype q. Then the following are equivalent.*

(1) *There are a Banach space Y of cotype q and a sequence $(y_n) \subset Y$ such that*

$$\mathcal{I} = \mathcal{B}_Y((y_n)) \quad \text{and} \quad \sum_{n=1}^{\infty} \|y_n\|_Y^q = \infty.$$

(2) *$\mathcal{I}$ is contained in a non-trivial summable ideal.*

Proof. Assume first that (1) holds. Proposition 3.7 of the source paper says that if Y has cotype q, then

$$\mathcal{B}_Y((y_n)) \subseteq \mathrm{Sum}((\|y_n\|_Y^q)_n).$$

Since $\sum_n \|y_n\|_Y^q = \infty$, the ideal $\mathrm{Sum}((\|y_n\|_Y^q)_n)$ is non-trivial. Thus $\mathcal{I}$ is contained in a non-trivial summable ideal.

Conversely assume that $\mathcal{I} \subseteq \mathrm{Sum}(w)$, where $w_n \ge 0$ and $\sum_n w_n = \infty$. Let

$$Y = X \oplus_q \ell_q, \qquad \|(u,a)\|_Y = (\|u\|_X^q + \|a\|_{\ell_q}^q)^{1/q},$$

and define $y_n = (x_n, w_n^{1/q} e_n)$.

The space Y has cotype q. Indeed, if X has cotype q with constant C_X, and ℓ_q has cotype q with constant C_q, then for every finite family $z_j = (u_j, v_j) \in Y$,

$$\begin{aligned}
\sum_j \|z_j\|_Y^q &= \sum_j \|u_j\|_X^q + \sum_j \|v_j\|_{\ell_q}^q \\
&\le C_X\, \mathbb{E}_\varepsilon \Big\| \sum_j \varepsilon_j u_j \Big\|_X^q + C_q\, \mathbb{E}_\varepsilon \Big\| \sum_j \varepsilon_j v_j \Big\|_{\ell_q}^q \\
&\le \max(C_X, C_q)\, \mathbb{E}_\varepsilon \Big\| \sum_j \varepsilon_j z_j \Big\|_Y^q .
\end{aligned}$$

For every finite $F \subseteq \mathbb{N}$, the definition of the ℓ_q-sum gives

$$\Big\| \sum_{n\in F} y_n \Big\|_Y^q = \Big\| \sum_{n\in F} x_n \Big\|_X^q + \sum_{n\in F} w_n.$$

Therefore, for every $A \subseteq \mathbb{N}$,

$$A \in \mathcal{B}_Y((y_n)) \iff A \in \mathcal{B}_X((x_n)) \text{ and } A \in \mathrm{Sum}(w).$$

Since $\mathcal{I} = \mathcal{B}_X((x_n)) \subseteq \mathrm{Sum}(w)$, this yields $\mathcal{B}_Y((y_n)) = \mathcal{I}$. Finally,

$$\sum_n \|y_n\|_Y^q = \sum_n (\|x_n\|_X^q + w_n) \ge \sum_n w_n = \infty.$$

This proves (1). ∎

Corollary 2. *Question 3.9 of arXiv:2501.15643 has an affirmative answer for every proper $\mathcal{B}$-ideal that is representable on a space of cotype q.*

Proof. The source paper recalls that $\mathcal{B}$-ideals are exactly the nonpathological F_σ ideals. Filipow–Tryba, *Path of pathology*, Theorem 11.2, proves that every nonpathological F_σ ideal is an intersection of summable ideals; in particular it is contained in a non-trivial summable ideal. Apply Theorem 1. □

Verification notes

The construction preserves the original cotype exponent q, not merely some larger finite cotype, because the added coordinate is ℓ_q and the direct sum is the ℓ_q-sum. The equality $\mathcal{B}_Y((y_n)) = \mathcal{B}_X((x_n)) \cap \mathrm{Sum}(w)$ is exact and uses only the finite-partial sum characterization of weak unconditional convergence.

The only external nontrivial input beyond the source paper is Filipow–Tryba's Theorem 11.2. It supplies the non-trivial summable ideal needed in Theorem 1. The theorem is stated for proper ideals, matching the convention in the source paper.

Novelty and limitations

I searched the run indexes for arXiv:2501.15643, the source title, cotype, $\mathcal{B}$-representability, and the summable-representation question. I also ran a web search for exact phrases from Question 3.9 and for the source/supporting paper titles. The search found the source paper and Filipow–Tryba's supporting paper, but no separate statement of the direct-sum cotype representation theorem above.

The answer relies on the standard characterization of $\mathcal{B}$-ideals as nonpathological F_σ ideals. If a reader uses a nonstandard convention allowing $\mathcal{P}(\mathbb{N})$ as an ideal, then the conclusion must exclude that improper case: $\mathcal{P}(\mathbb{N})$ can be represented by the zero sequence, but it cannot be represented in cotype q with divergent $\sum \|x_n\|^q$, because Proposition 3.7 would force $\mathbb{N} \in \mathrm{Sum}((\|x_n\|^q))$.

Human review recommendation

This is a short candidate full solution. The main review points are: verify the convention that all ideals in the source question are proper; verify the cited equivalence between $\mathcal{B}$-ideals and nonpathological F_σ ideals; and check that the ℓ_q-sum cotype estimate is acceptable with the source paper's normalization of cotype constants.

References

[1] J. Lopez-Abad, V. Olmos-Prieto, and C. Uzcategui-Aylwin, *F_σ-ideals, colorings, and representation in Banach spaces*, arXiv:2501.15643.
[2] R. Filipow and J. Tryba, *Path of pathology*, arXiv:2501.00503.